## SBL (Sparse Bayesian Learning)

$x \sim N(0, P)$ 先验 Prior

$y|x \sim N(Ax, \sigma^2 I)$

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)} \propto p(y|x)p(x)$$

Gauss RV 的和为 Gauss

Gauss PDF 的积为 Gauss

$x|y \sim N(\mu_{x|Y}, \Sigma_{x|Y})$

$\mu_{x|Y} = PA^H R^{-1} y \leftarrow \hat{x}_{MMSE}$

$\Sigma_{x|Y} = P - PA^H R^{-1} AP$

## Off-grid Direction of Arrival Estimation Using Sparse Bayesian Interference

$$\begin{tikzcd} & X & \Lambda \arrow[l, "N"'] & \rho \arrow[l, "\Gamma"'] \\ Y \arrow[ur] & & & \\ & E \arrow[ul] & \beta \arrow[l, "N"'] & c.d \arrow[l, "\Gamma"'] \end{tikzcd}$$

$X \sim N(0, \Lambda)$

$E \sim N(0, \beta^{-1} I)$

$X|Y, P, \sigma^2 \sim N(\mu, \Sigma)$

$\mu = \beta \Sigma A^H y$

$\Sigma = (\beta A^H A + \Lambda^{-1})^{-1}$

$= \Lambda - \Lambda A^H R^{-1} A \Lambda$ 和上边一样

SBL + EM

$\gamma, \beta$ 可由 EM 确定

$\Lambda, \beta \leftarrow X$

$X \leftarrow \Lambda, \beta$